Cogtics – Comprehensive Backend Plan

Production-Ready Architecture, Data Model, and Execution Strategy

1. Backend North Star

Cogtics ingests data from APIs and manual uploads, normalizes it into a canonical truth model, snapshots system state, computes revenue recognition, runs deterministic detection rules, generates evidence packs, supports simulations, and exports audit-grade outputs.

2. Architecture Blueprint

The system is built as a modular monolith with clear internal boundaries covering ingestion, normalization, identity resolution, contracts, revenue recognition, detection, simulation, exports, and audit.

3. Canonical Data Model

All computations run against immutable snapshots. Snapshots ensure replayability, auditability, and deterministic outputs.

4. Core Data Entities

Key entities include tenants, users, connections, sync runs, canonical customers, contracts, contract terms, invoices, revenue obligations, revenue schedules, issues, simulations, exports, and audit logs.

5. Connector Framework

Each connector implements OAuth authentication, incremental sync, schema discovery, retries, rate-limit handling, and health checks. Connectors are versioned independently.

6. Ingestion Pipeline

The ingestion pipeline fetches, validates, normalizes, resolves identity, commits canonical data, builds snapshots, and triggers downstream processing.

7. Identity Resolution

CRM systems anchor customer identity. Billing and accounting records are matched using deterministic and assisted rules, with human confirmation for ambiguities.

8. Contract Intelligence

Contracts are ingested as PDFs, parsed into clause-level terms with confidence scores, versioned, and gated based on readiness thresholds.

9. Revenue Recognition Engine

The rev-rec engine supports performance obligations, allocation, straight-line schedules, recognized versus deferred revenue tracking, and reconciliation.

10. Detection Engine

Deterministic rules detect revenue leakage and recognition mismatches. Issues include expected versus actual calculations, confidence, severity, and evidence references.

11. Evidence Generation

Evidence packs are immutable and include calculation tables, contract clauses, and source record references.

12. Simulation Engine

Simulations clone snapshots and apply overrides to compute deltas without mutating source data. Multiple fixes and reusable templates are supported.

13. Exports and Digests

Exports include CSV, PDF, and email digests. All exports are point-in-time, versioned, and audit-ready.

14. Security and Compliance

Tenant isolation, encryption, secrets management, and audit logging are enforced system-wide.

15. Observability and Recovery

Monitoring, retries, dead-letter queues, snapshot replay, and feature flags ensure reliability and safe operations.

16. Build Order

The backend is built in stages covering foundations, connectors, canonical model, contracts, revenue recognition, detection, simulation, exports, and hardening.

17. Production Readiness Criteria

A tenant is production-ready when ingestion is resilient, detections are reproducible, simulations are isolated, exports are audit-grade, and all actions are logged.